Project Report

Fundamentals of Software Project Management

FOODHUB - ONLINE FOOD ORDERING & DELIVERY MANAGEMENT SYSTEM

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Business Statement

Organization Background

Alvigha is an established restaurant chain operating multiple successful locations across Karachi. Known for their signature dishes and high-quality food offerings, Alvigha has built a strong reputation and loyal customer base over the years. Their existing branches have consistently demonstrated strong dine-in customer traffic and positive customer satisfaction ratings.

With the rapid growth in food delivery markets and changing consumer behaviors, Alvigha has identified a strategic opportunity to expand their operations by creating a new delivery app that covers major districts with high demand for food delivery services. The area demographics show a predominantly working-class population with busy lifestyles, making it an ideal location for delivery-focused food service operations.

Currently, Alvigha relies heavily on third-party delivery platforms such as Foodpanda, Cheetay, and Uber Eats for their delivery operations. While these platforms provide market access, they come with significant commission fees (ranging from 20-35% per order), limited control over customer experience, and minimal access to customer data for building direct relationships.

Opportunity

The Delivery Management System presents a unique opportunity due to several favorable market conditions:

1. **High Delivery Demand:** Market research indicates that 68% of residents in Gulshan order food delivery at least 3-4 times per week, with peak ordering times during lunch hours (12 PM - 2 PM) and dinner hours (7 PM - 10 PM).
2. **Limited Direct-to-Consumer Competition:** Most restaurants in the area rely exclusively on third-party platforms, with very few offering their own branded delivery solutions.
3. **Brand Recognition:** Alvigha's signature dishes are already well-known in the city, creating immediate market awareness for the new location.
4. **Digital-First Demographics:** The target customer base (ages 22-45) demonstrates high smartphone penetration and comfort with mobile app-based ordering.

Problem

Despite the strong market opportunity, Alvigha faces several critical challenges:

Challenge	Description/Impact	Quantitative Effect
High Commission Costs	Third-party platforms charge substantial fees per order, directly reducing net profit.	Commission rates of 25-30% per order. Leads to substantial revenue loss on 3,000–5,000 projected monthly orders.
Loss of Customer Relationships	No direct access to customer data, preferences, or ordering history.	Impossible to build loyalty programs or conduct targeted marketing.
Brand Dilution	Competition against dozens of other restaurants on multi-platform apps.	Dilutes brand visibility and reduces the ability to showcase unique value propositions.
Limited Control Over CX	Delivery times, rider behavior, food handling, and customer service are controlled by third-party platforms.	Leads to inconsistent quality and potential brand reputation damage.
Competitive Disadvantage	Customers lack incentive to choose Alvigha over competitors on the same platform.	Pricing and delivery fees are standardized across platforms.
Lack of Operational Insights	Third-party platforms provide limited analytics.	Restricts the ability to optimize operations based on customer behavior, peak times, popular items, and delivery efficiency.

Project Team

Core Project Team Structure

Role	Headcount	Key Responsibilities	Justification for Resource
Project Manager	1	Overall project coordination, timeline management, risk mitigation, and quality assurance oversight.	Essential for coordinating multiple development streams and ensuring project delivery within the 6-month timeframe.
Business Analyst	1	Requirements gathering, process flow mapping, user story creation, and validation of stakeholder requirements.	Required to bridge the gap between business objectives and technical implementation to ensure the solution meets operational needs.
Frontend Developers	2	Mobile app and web portal development, UI component implementation, API integration, and cross-platform compatibility testing.	Needed for parallel development of the Customer Mobile App (iOS/Android) and the Restaurant Web Portal using modern frameworks.
Backend Developers	2	RESTful API development, database design and optimization, order processing logic, payment gateway integration, and real-time tracking implementation.	Essential for building a robust server-side architecture capable of handling concurrent orders, real-time updates, and third-party system integrations.
UI/UX Designers	2	User research, wireframing, high-fidelity mockup creation, user flow design, and brand identity design.	Necessary to create user-friendly interfaces for four distinct user types (customers, riders, staff, admin) to ensure the system is competitive.
QA Tester	1	Test case creation, functional, integration, performance, and security testing, plus bug documentation and UAT coordination.	Dedicated resource to ensure system reliability and validate business logic across platforms, mitigating immediate business impact from failures.
Database Administrator	1	Database design and normalization, performance tuning, data security implementation, and automated backup/recovery procedures.	Required to manage sensitive customer and payment data, ensuring data integrity and optimizing query performance under load.

Project Team Organizational Structure - Controlled Decentralized Model

The FoodHub project employs a **Controlled Decentralized** organizational structure that balances autonomy with coordination. Each functional area operates independently under a designated leader who has decision-making authority within their domain, while the Project Manager provides overall coordination, ensures alignment with project objectives, and facilitates cross-functional collaboration.

```
Project Sponsor (Management)
                ↓
 Project Manager (Coordination)
                ↓
        _____|_____
     ↓                 ↓
Technical Lead      Quality Lead
(Autonomous Departments)
```

Resource Identification – Expert Opinion (Muhammad Ali)

"The selected roles are essential for delivering a scalable, user-focused platform. Each resource supports a critical project function - from requirements and design to development, integrations, testing, and database management - ensuring smooth coordination across the Controlled Decentralized structure."

Feasibility Analysis

Technical Feasibility Analysis (Rating: 9/10)

Category	Details/Status	Key Technologies
Assessment & Justification	The platform is **highly feasible** using proven, mature, industry-standard technologies. The minor risks associated with high load are manageable using standard engineering practices.	N/A

Technical Stack	A robust, scalable, and modern stack covering all layers of the application.	**Mobile Apps:** React Native/Flutter, **Backend:** Node.js/Django/Python, **Database:** PostgreSQL/Redis, **Cloud:** AWS/Google Cloud.
Positive Factors (Strengths)	* **Proven Technologies:** Extensive documentation, large developer communities, and established scalability. * **Team Expertise:** Developers cover all required skills (mobile, backend, database). * **Third-Party Integration:** APIs for payments/mapping are well-documented, reducing custom work. * **Scalable Architecture:** Cloud hosting supports scaling from hundreds to thousands of concurrent orders.	N/A
Challenges (Risks)	* **Real-Time Tracking:** Accurate, simultaneous GPS tracking requires careful performance optimization. * **Concurrent Order Handling:** Managing 100+ simultaneous orders during peak hours requires robust architecture and load testing.	**Real-Time Comm:** WebSocket, Socket.io, **Notifications:** FCM, Twilio.
Mitigation Strategies	* Implement comprehensive **load testing** before launch. * Design for **horizontal scalability** from day one. * Use proven real-time technologies (WebSocket/Firebase). * Prioritize performance optimization with experienced developers.	N/A

Economic Feasibility Analysis (Rating: 8/10)

Category	Cost/Metric	Details
I. Project Investment (6 Months)	**PKR 5,060,000** (Approx. $18,000)	Total initial cost, including a 15% contingency.
Development Team Salaries	PKR 3,500,000	Core technical team salaries.
Infrastructure & Hosting (Initial)	PKR 300,000	Setup for cloud services.

Third-Party Services (Licenses)	PKR 200,000	APIs and necessary licenses.
Operational Resources (Equipment)	PKR 400,000	Delivery bags, uniforms, etc.
Contingency (15%)	PKR 660,000	Buffer for unforeseen costs.
II. Ongoing Operational Costs (Monthly)	**PKR 185,000 + Fees**	Fixed monthly expenses post-launch.
Cloud Hosting & Infrastructure	PKR 50,000	Ongoing cloud usage.
SMS/Push Notification Services	PKR 20,000	Ongoing customer communication costs.
Google Maps API Usage	PKR 15,000	Ongoing location services.
Maintenance & Support	PKR 100,000	Team/vendor maintenance.
Payment Gateway Fees	Variable (Approx. PKR 90,000)	2.5% of projected transaction value.
III. Revenue & ROI Projection	**PKR 1,085,000**	Estimated Monthly Net Gain from owning the platform.
Current Monthly Net Revenue	PKR 2,240,000	Revenue after losing PKR 960,000 (30% commission).
Future Monthly Net Revenue	PKR 3,325,000	Revenue after increased orders and reduced costs.
Payback Period	**4.7 months**	Time required to recoup the initial PKR 5,060,000 investment.
Annual ROI	**257%**	Strong return justifying the upfront investment.
IV. Challenges & Mitigation	N/A	Minor risks are manageable, ensuring financial viability.
Main Challenges	Upfront capital requirement; dependency on market adoption rate.	N/A

Mitigation	Phased investment approach; aggressive customer acquisition campaigns and promotions.	N/A

Operational Feasibility Analysis (Rating: 7/10)

Category	Details/Status	Key Operational Requirements
Assessment & Justification	The platform is **operationally feasible** but requires significant effort in organizational change, staff training, and process redesign. Success hinges on effective change management.	N/A
Operational Requirements	Detailed changes needed across the organization.	**Staff Training:** Adapt to digital order management and use admin/analytics tools. **Process Changes:** Transition from phone/verbal orders to app-based ordering, implement real-time inventory management, establish 15-20 min order preparation **SLAs**.
Infrastructure Requirements	Physical resources needed for core functionality.	Stable internet connectivity, tablets/computers for staff order monitoring, smartphones for riders, and backup power/charging stations.
Positive Factors (Strengths)	* **Management Support:** Top-down commitment to operational change. * **Existing Operations:** Staff familiarity with takeout orders, food packaging, and timing. * **Phased Rollout:** Ability to launch with limited delivery zones to reduce complexity.	N/A

Challenges (Risks)	* **Staff Adoption:** Resistance to new digital processes. * **Quality Control:** Need to establish new internal processes for consistent delivery times and food quality. * **Customer Service:** Requirement to build a dedicated in-house support team from scratch.	N/A
Mitigation Strategies	* **Training:** Implement comprehensive **2-week intensive training** program. * **Pilot Phase:** Launch with limited capacity to allow staff adjustment. * **Support Team:** Hire 2 dedicated customer support representatives. * **Incentives:** Use performance bonuses and feedback loops to drive adoption and quality.	N/A

Legal Feasibility Analysis (Rating: 8/10)

Category	Key Requirements	Challenges & Mitigation
Assessment & Justification	The project is **highly legally feasible** as all required compliance frameworks are established in Pakistan and are well-documented.	N/A
Data Protection & Privacy	Compliance with Pakistan's data protection regulations, including secure storage of customer personal information.	**Challenge:** Data Breach Liability due to storing customer data (e.g., payment info). **Mitigation:** Comprehensive security audits and insurance.
Payment Processing	Compliance with State Bank of Pakistan regulations, **PCI-DSS Level 1 compliance** for	**Challenge:** Achieving and maintaining PCI-DSS compliance requires ongoing

	credit card handling, AML requirements, and transaction audit trails.	audits and infrastructure investment. **Mitigation:** Use certified third-party payment gateways to offload burden.
Labor & Employment	Defining delivery rider contracts (employee vs. contractor classification), ensuring minimum wage compliance, and securing insurance/liability coverage.	**Challenge:** Rider Classification impacts tax obligations and liability. **Mitigation:** Expert consultation to establish clear contractual agreements.
Intellectual Property (IP)	Trademark registration for the "FoodHub" brand, copyright protection for app code/designs, and adherence to open-source license compliance.	N/A
Third-Party Agreements	Executing contracts with payment gateways, defining Service Level Agreements (SLAs) with cloud hosting providers, and securing API licensing (e.g., Google Maps).	N/A
Positive Factors	Established legal frameworks in Pakistan; third-party partners (gateways, cloud) maintain existing compliance certifications, reducing the burden on Alvigha.	N/A

Critical Success Factors (CSFs)

Critical Success Factors are the essential elements that must go right for the FoodHub project to achieve its objectives. These factors represent areas requiring focused attention, measurement, and management throughout the project lifecycle.

CSF 1: User Adoption Rate

Description: Achieving target customer adoption rates is critical for ROI realization. The success of FoodHub depends on successfully migrating customers from third-party platforms to the Alvigha app.

Why Critical:

- Revenue projections assume 4,500 monthly orders through FoodHub
- Low adoption means continued dependency on third-party platforms
- Network effects: more users attract more riders, improving service quality

CSF 2: System Reliability and Performance

Description: The platform must maintain 99.5%+ uptime and handle peak loads without performance degradation. System failures during meal times directly translate to lost revenue and customer dissatisfaction.

Why Critical:

- Peak ordering times (12-2 PM, 7-10 PM) represent 70% of daily revenue
- A single hour of downtime during peak time = ~PKR 40,000 lost revenue
- Customer trust is fragile; repeated failures will drive users back to competitors
- Rider productivity depends on reliable app performance

Measures Taken:

- Comprehensive load testing simulating 200+ concurrent orders
- Redundant server infrastructure with automatic failover
- Real-time monitoring and alerting for system health
- 24/7 technical support during first 3 months
- Gradual rollout starting with limited delivery zones
- Target KPIs: 99.5% uptime, <2 second API response time, <3% order failure rate

CSF 3: Delivery Service Quality

Description: Consistent, timely, and professional delivery service is essential for customer satisfaction and competitive advantage. Poor delivery experiences will drive customers back to established platforms.

Why Critical:

- 80% of customer satisfaction in food delivery depends on timely, intact delivery
- Google/Facebook reviews heavily influence new customer acquisition
- Delivery time is the top differentiator between food delivery platforms
- Poor delivery experiences have higher viral spread (negative word-of-mouth) than positive ones

Measures Taken:

- Rigorous rider recruitment with background checks and interviews
- Mandatory 3-day training program covering app usage, customer service, and safety
- GPS-based route optimization to minimize delivery times
- Real-time rider performance monitoring and feedback
- Quality control checks: insulated bags, vehicle cleanliness, uniform standards
- Customer rating system for riders with performance-based incentives
- Target KPIs: 30-minute average delivery time, 4.5+ star average rider rating, <5% late deliveries

CSF 4: Restaurant Operational Integration

Description: Seamless integration of the FoodHub system with existing restaurant kitchen and front-of-house operations is essential for order fulfillment accuracy and speed.

Why Critical:

- Kitchen workflow disruptions cause delays affecting all customers (dine-in and delivery)
- Order errors (wrong items, missing items) are the #2 cause of negative reviews
- Staff resistance to new systems can sabotage adoption
- Inefficient workflows increase order preparation time, reducing delivery capacity

Measures Taken:

- 2-week comprehensive staff training before launch
- Dedicated tablets for kitchen order display with audio alerts
- Simplified order management interface designed with staff input
- Real-time inventory sync to prevent orders for out-of-stock items
- "Ready for pickup" notification system to optimize rider dispatch
- Pilot period with reduced order volume to allow staff adjustment
- Regular feedback sessions with kitchen and front-desk staff
- Target KPIs: 15-minute average order preparation time, <2% order error rate, 90%+ staff satisfaction

CSF 5: Budget and Timeline Adherence

Description: Delivering the FoodHub platform within the 6-month timeline and PKR 5.06 million budget is critical for project viability and management confidence.

Why Critical:

- Launch timing aligned with new branch opening (significant marketing investment at stake)
- Budget overruns may jeopardize project approval or require compromise on features
- Delays beyond 6 months increase opportunity cost (continued third-party commissions)
- Management support depends on demonstrated project management competence

Measures Taken:

- Detailed project plan with 180+ activities and critical path analysis
- Weekly progress reviews and milestone tracking
- 15% contingency budget allocation for unforeseen issues
- Agile development methodology with 2-week sprints for flexibility
- Early identification and mitigation of risks
- Clearly defined scope with change control process to prevent scope creep
- Prioritized feature list with MVP (Minimum Viable Product) definition
- Target KPIs: 100% milestone completion on time, <5% budget variance

Project Goal/Purpose/Aim/Objective

Project Aim

The overarching aim of the FoodHub project is to establish Alvigha as an independent, customer-centric food delivery service provider in Gulshan by developing a comprehensive, branded online ordering and delivery management platform that eliminates third-party dependencies, maximizes profitability, and delivers superior customer experiences.

This strategic initiative will position Alvigha for sustainable growth in the rapidly expanding food delivery market while maintaining full control over customer relationships, operational workflows, and brand presentation.

Project Goals and Success Metrics

Goal	Strategic Objective	Key Success Metric
1. Eliminate Third-Party Dependency	Completely transition the majority of delivery orders to the proprietary FoodHub platform.	≥90% of delivery orders processed through FoodHub, <10% through third-party platforms by Month 6.
2. Maximize Profit Margins	Increase net profit margins by eliminating high third-party commission fees and optimizing logistics.	Net profit margin on delivery orders increases from current 15–20% to 45–50% within 3 months of launch.
3. Establish Direct Customer Relationships	Build a proprietary customer database to enable targeted marketing and loyalty programs.	5,000+ registered users, 2,500+ monthly active users by Month 12.
4. Enhance Brand Visibility and Control	Create a distinctive, branded digital presence showcasing unique quality standards without external competition.	Brand recall increased by 40%, 4.5+ star average app rating on app stores.
5. Deliver Superior Customer Experience	Provide faster, more reliable, and personalized service to drive satisfaction and repeat orders.	Average delivery time 25% faster than third-party platforms, NPS ≥70, 60%+ repeat order rate.

Project Objectives (SMART)

Objective	Description	Target / Success Metric	Due Date / Timeline
1. Platform Development and Deployment	Develop and deploy a fully functional, multi-platform food ordering and delivery management system.	Includes Customer Apps (iOS/Android), Rider App, Staff Web Portal, and Admin Dashboard.	Complete deployment by **May 31, 2026** (6 months from project start).
2. Achieve Target User Base	Acquire registered users through aggressive marketing campaigns and promotions.	**2,000** registered users by Month 3; **5,000** registered users by Month 12.	Within the first 12 months post-launch.
3. Process Target Order Volume	Successfully process a defined volume of food delivery orders through the FoodHub platform.	Process **3,000+** monthly orders by Month 2, scaling to **5,000+** monthly orders by Month 6.	Within 6 months of operation.
4. Maintain Service Quality Standards	Achieve and consistently maintain high customer satisfaction and quality delivery metrics.	Minimum **4.3-star** average app rating; <5% order complaints; <3% late deliveries.	Maintained consistently post-launch.
5. Achieve System Reliability	Ensure high platform uptime and fast system performance, especially during peak periods.	≥99.5% uptime; average API response time **<2 seconds**; zero critical failures during peak hours (12 PM-2 PM, 7 PM-10 PM).	Maintained consistently post-launch.
6. Realize Financial Returns	Achieve project ROI breakeven by eliminating commissions and increasing order volume.	Cumulative savings exceed total project investment by **Month 5** post-launch.	Within 5 months of launch.

Benefits to the Organization

Financial Benefits:

1. **Increased Revenue:** Projected 12.5% increase in delivery order volume due to promotional incentives and improved customer experience (from 4,000 to 4,500 monthly orders).
2. **Improved Profit Margins:** Increase net profit margin on delivery orders from 15-20% to 45-50%, adding approximately PKR 1,085,000 in monthly profit.
3. **Reduced Commission Costs:** Eliminate PKR 960,000 in monthly third-party platform commission fees.
4. **Rapid ROI:** Recover full project investment within 4.7 months of launch, achieving 257% annual ROI.
5. **Lower Customer Acquisition Costs:** Direct marketing to owned customer base reduces dependency on expensive third-party advertising.

Customer Experience Benefits:

1. **Faster Delivery Times:** Optimized routing and dedicated fleet targeting 25% faster delivery compared to third-party platforms (30-minute average vs. 40-45 minutes).
2. **Personalized Experience:** Customer order history, preferences, and personalized recommendations create superior user experience.
3. **Loyalty Rewards:** Proprietary loyalty program encourages repeat orders and builds long-term customer relationships.
4. **Better Customer Support:** Dedicated support team familiar with Alvigha operations provides faster, more empathetic complaint resolution.
5. **Exclusive Benefits:** App-only menu items, special discounts, and early access to new dishes create differentiation and value.

Project Deliverables and Success Criteria

ID	Deliverable	Key Features/Components	Linked Objectives	Success & Acceptance Criteria (Highlights)
1	**Customer Mobile App** (iOS & Android)	User authentication (social/email/phone), real-time order tracking, menu browsing, multiple payment options (COD, card, wallet), loyalty points, ratings, and push notifications.	Obj 1, 2, 3	**Success:** Available on App Stores by May 31, 2026; 4.3+ star rating; App crashes <0.1%. **Acceptance:** Load testing 500+ concurrent users; Security audit completed.
2	**Delivery Rider App** (Android)	Real-time order assignment, turn-by-turn navigation, delivery confirmation (PIN/signature), earnings tracking, availability toggle, and SOS button.	Obj 1, 5	**Success:** GPS accuracy within 10m; Works in low-connectivity (offline capability); 8+ hours battery optimization. **Acceptance:** Tested on 5$+$ device models; Integration with order management verified.
3	**Restaurant Staff Web Portal**	Real-time order queue with audio alerts, KDS mode, menu management, order status updates, real-time inventory toggles, and staff account management.	Obj 1, 4	**Success:** Accessible on modern browsers; Order update time <5 seconds; 100% staff training completion. **Acceptance:** UAT completed with 5+ staff; Performance testing handles peak orders.
4	**Administrative Dashboard**	Real-time operational dashboard, comprehensive analytics/reporting, user management, promotional campaign management, financial reconciliation, and data export.	Obj 6, 7	**Success:** Key metrics updated in real-time (<30 sec lag); 100% data accuracy; Accessible via secure authentication/audit logs. **Acceptance:** Management trained; Security penetration testing completed.
5	**Backend API & Infrastructure**	RESTful API services, PostgreSQL DB with Redis caching, real-time WebSocket services, integrations (Payment, Google Maps, SMS/Push), and security (JWT, OAuth).	Obj 1, 6	**Success:** API response time <2 seconds; Uptime 99.5%+; Automated backup and recovery procedures. **Acceptance:** Load testing 500$+$ concurrent users; API documentation complete (Swagger/OpenAPI).

6	**Cloud Infrastructure Setup**	Load-balanced application servers (auto-scaling), managed DB services, Content Delivery Network (CDN), SSL/HTTPS configuration, and CI/CD pipeline setup.	Obj 6, 7	**Success:** Supports 10× projected load; Auto-scaling tested; Disaster Recovery RTO <1 hour. **Acceptance:** Infrastructure-as-code documentation created; Security audit passed.
7	**Delivery Operations Resources**	30 insulated, branded delivery bags, 20 branded rider uniforms, 20 smartphone mounts, power banks, vehicle branding materials, and safety equipment.	Obj 4, 5	**Success:** Equipment delivered 2 weeks before launch; Materials meet brand guidelines. **Acceptance:** Physical materials inspected and approved; Vendor contracts secured.
8	**Training and Documentation**	User manuals (Staff/Admin/Rider), video tutorials, quick reference guides, 2-day staff workshop, and technical documentation (architecture, API, deployment).	Obj 4, 5	**Success:** 100% staff training completion before launch; 85%+ post-training comprehension validated. **Acceptance:** All training materials approved; Documentation accessible in centralized repository.
9	**Marketing and Launch Materials**	Brand identity guidelines, social media content, launch promotional campaign (25% off), referral program materials, in-store promotion materials, and digital ad creatives.	Obj 2, 4	**Success:** 500+ app downloads in first week; 50,000+ social media impressions. **Acceptance:** Marketing materials approved; Campaign timeline and budget finalized.
10	**Quality Assurance & Testing Reports**	Functional, performance, security (penetration testing), usability, and integration testing reports; UAT report with stakeholder sign-offs.	Obj 1, 6	**Success:** Zero critical bugs at launch; Performance benchmarks met; All security vulnerabilities addressed. **Acceptance:** All test plans executed; Stakeholder sign-off on UAT results.

Project Scope

Project Boundaries

The FoodHub project is bounded by a clear focus on establishing a food delivery platform for Alvigha's. The project encompasses the complete technology stack (mobile applications,

web portals, and backend infrastructure), operational setup for a delivery fleet serving a 5km radius, and launch marketing initiatives. This is a greenfield implementation designed to replace third-party platform dependency with a proprietary system, covering the full customer journey from order placement to delivery completion. The project prioritizes core delivery functionality and proven technologies over experimental features, focusing on rapid deployment within six months. While the technical architecture is designed for future scalability, the initial scope deliberately limits implementation to a single location to ensure operational excellence and system stability before expansion. Advanced features such as multi-location management, B2B services, AI-powered enhancements, and third-party integrations are intentionally deferred to future phases, allowing the team to deliver a robust, well-tested foundation that addresses the immediate business need of capturing the Gulshan delivery market.

Things Included in Project Scope

Category	Key Features / Functionality	Technical Details
1. Customer-Facing App	Ordering & Browsing: Complete menu browsing (with images, prices, nutritional info), advanced search/filtering, customization options (spice, add-ons), cart management, loyalty points, and promotional coupon application.	Engagement: Push notifications, rating/review system, in-app customer support (chat/call/email), first-time user onboarding.
2. Rider-Facing App	Logistics & Dispatch: Real-time order assignment, turn-by-turn navigation, availability toggle (online/offline), order acceptance/rejection (with time limits).	Management: Rider profile/document management, earnings dashboard (real-time tracking, weekly statements), delivery confirmation (PIN/signature), SOS/emergency support button.
3. Restaurant Staff Portal	Order Management: Real-time order queue display (with audio alerts), order status updates (received → preparing → ready → picked up).	Content Control: Menu management (CRUD), category management, real-time inventory toggles (out-of-stock), Kitchen Display System (KDS) mode, daily sales summary/reports.

4. Administrative Features	Monitoring: Real-time operational dashboard (orders, riders, revenue), analytics dashboards, audit logs.	Management Tools: User/staff/rider management, delivery zone configuration, pricing management, promotional campaign manager, financial reports, data export (CSV, Excel, PDF).
5. Technical Infrastructure	Scalability: Scalable cloud infrastructure with auto-scaling, load balancing, database replication/automated backups, CDN.	Core Systems: Real-time WebSocket connections, scheduled jobs (payment processing, reports), logging, monitoring, and error tracking/alerting.
6. Integration Services	Payments: Stripe, JazzCash, EasyPaisa, Cash on Delivery.	Geo/Comms: Google Maps API (geocoding, routing), SMS (Twilio/local), Push Notifications (FCM/APNs), Email (SendGrid/AWS SES), Analytics (Google Analytics).
7. Security Features	Authentication: Secure user authentication (email/password, OTP), JWT token-based API authentication, bcrypt password encryption, Role-Based Access Control (RBAC).	Compliance: PCI-DSS compliant payment processing (tokenization), HTTPS encryption, activity audit logs, data privacy compliance, API request throttling.
8. Marketing & Launch	Campaigns: Launch campaign (25% off first 3 orders), referral program (PKR 100 credit), loyalty program mechanics, social media marketing.	Assets: Google Ads/Facebook Ads setup, in-store promotional materials, email marketing for launch announcements.

Things Excluded from Project Scope

Category of Exclusion	Excluded Features and Rationale	Rationale/Future Phase
1. Multi-Location Support	Centralized management for multiple restaurant locations; Inter-branch order routing or inventory sharing.	Platform will serve **ONE branch only** (Gulshan location).
2. Third-Party Integration	Listing other restaurants on the platform; Marketplace functionality.	FoodHub is strictly proprietary and will **only serve Alvigha**.

3. Advanced/AI Features	AI-powered chatbot support; Voice ordering; AR menu visualization; Blockchain/Cryptocurrency payments; Advanced ML fraud detection or demand forecasting.	Deferred to **Future Phases** to focus resources on the Minimum Viable Product (MVP).
4. POS Integration	Direct electronic integration with the existing restaurant POS system.	Orders will be manually entered by staff. Integration is deferred to **Phase 2**.
5. Dine-In Services	Table reservation system; Dine-in menu ordering from tables; QR code table ordering.	Project focus is **100% on delivery** operations.
6. Geographic Coverage	Expansion to other parts of the city.	Delivery is strictly **limited to a 5km radius** from the Gulshan branch.
7. Corporate/B2B Features	Bulk ordering for offices; Corporate account management; Invoice/PO-based billing.	Focus is exclusively on the consumer (B2C) market in Phase 1.
8. Subscription Services	Meal subscription plans; Meal prep services; Unlimited delivery subscriptions.	Feature set limited to single, on-demand orders.
9. Advanced Delivery	Drone delivery; Autonomous vehicle delivery; Scheduled delivery or time slots.	Delivery model is limited to **ASAP (as soon as possible)** courier service.
10. Social/Community	User profiles visible to others; Social sharing of orders; Group ordering functionality; Gamification beyond loyalty points.	Focus is on transaction efficiency and privacy.

Project Phases

The FoodHub project is structured into 5 distinct phases over the 6-month timeline:

Phase 1: Initiation & Planning (Weeks 1-2) Duration: December 1-14, 2025

Activities:

• Project kickoff meeting with all stakeholders	• Requirements gathering and documentation	• Expert consultation sessions

• Team formation and role assignments	• Development environment setup	• Design sprint and wireframing
• Technical architecture finalization	• Vendor selection (payment gateway, SMS provider, cloud hosting)	• Project plan finalization and approval

Deliverables:

- Detailed requirements document
- System architecture diagram
- UI/UX wireframes and mockups
- Project plan with WBS and Gantt chart
- Risk management plan

Phase 2: Design & Development (Weeks 3-16) Duration: December 15, 2025 - March 29, 2026

Activities:

Duration	Focus	Key Activities
Weeks 3-6	**Foundation**	Database schema design and implementation; Backend API architecture setup; Authentication and authorization implementation; Payment gateway integration; Admin dashboard development; Initial mobile app UI implementation.
Weeks 7-10	**Core Features**	Customer app development (menu browsing, cart, checkout); Rider app development (order assignment, navigation); Restaurant portal development (order management); Real-time order tracking implementation; Notification services setup; Google Maps integration.
Weeks 11-14	**Advanced Features**	Loyalty program implementation; Ratings and reviews system; Analytics and reporting dashboards; Order history and reordering; Promotional code system; Customer support features.

Weeks 15-16	**Integration & Polish**	Full system integration testing; UI/UX refinements; Performance optimization; Bug fixing; Documentation.

Deliverables:

- Functional customer mobile apps (iOS & Android)
- Functional rider mobile app (Android)
- Restaurant staff web portal
- Administrative dashboard
- Backend APIs and database
- Integrated third-party services

Phase 3: Testing & Quality Assurance (Weeks 17-20) Duration: March 30 - April 26, 2026

Activities:

- Unit testing of all components
- Integration testing across systems
- Functional testing of all features
- Performance and load testing (simulate 500+ concurrent users)
- Security testing and penetration testing
- Usability testing with real users
- Mobile device compatibility testing (10+ devices)
- User acceptance testing (UAT) with stakeholders
- Bug fixing and regression testing
- Documentation review and updates

Deliverables:

- Test reports (functional, performance, security, usability)
- Bug tracking and resolution log
- UAT sign-off from stakeholders
- Updated technical documentation

Phase 4: Deployment & Operational Setup (Weeks 21-22) Duration: April 27 - May 10, 2026

Activities:

Category	Key Activities

I. Technical Deployment	Production cloud infrastructure setup; Database migration and optimization; SSL certificate installation and security configuration.
II. App Store Release	App submission to iOS App Store and Google Play Store; App store optimization (ASO); App approval and publication.
III. Training & Readiness	Restaurant staff training (2-day workshop); Rider training (3-day program); Administrative user training.
IV. Operational Setup	Delivery operations setup (zone mapping, pricing configuration); Delivery equipment procurement and distribution.
V. Final Checks	Marketing campaign preparation; Go-live checklist verification.

Deliverables:

- Live production system on cloud infrastructure
- Published mobile apps on app stores
- Trained restaurant staff (100% completion rate)
- Trained delivery rider fleet (20 riders)
- Delivery operational resources distributed
- Launch marketing materials ready

Phase 5: Launch & Stabilization (Weeks 23-26) Duration: May 11 - June 7, 2026

Activities:

Category	Key Activities
I. Initial Launch & Marketing	Soft launch (limited user base, single delivery zone); Launch marketing campaign activation; Social media promotion and influencer partnerships.
II. Monitoring & Technical Support	Monitoring system performance and stability; Immediate bug fixes and issue resolution; 24/7 technical support availability.

III. Optimization & Feedback	Customer feedback collection and analysis; Operational workflow adjustments; Daily stand-up meetings for issue tracking.
IV. Expansion & Review	Progressive rollout (expanding delivery zones); Full public launch (Week 24); Post-launch review and optimization.
V. Performance Management	Rider performance monitoring and coaching.

Deliverables:

- Live platform serving customers in Gulshan
- Active user base (target: 2,000+ registered users)
- Monthly order volume (target: 3,000+ orders by Month 2)
- Post-launch review report
- Lessons learned documentation
- Handover to ongoing operations team

Success Criteria for Project Scope Delivery

Category	Criterion	Target / Metric
I. Functional Completeness	All included features (as defined in the Scope document) implemented and operational.	Zero critical defects in production. Less than 10 minor defects with documented workarounds.
II. Performance Standards	System uptime (availability)	99.5%+ during first month post-launch.
	API response time	<2 seconds (95th percentile).
	Mobile app stability	Crash rate <0.1% of sessions.

	Order completion rate	>95% successful order completion.
III. User Adoption	Registered user base	2,000+ registered users within 3 months of launch.
	Order volume	3,000+ monthly orders by Month 2 post-launch.
	Customer satisfaction	4.3+ star average rating on app stores.
IV. Operational Readiness	Staff training completion	100% restaurant staff training completion.
	Delivery fleet availability	20 trained delivery riders operational.
	Equipment availability	All delivery equipment procured and distributed.
	Process adherence	SOPs documented and followed.
V. Quality Validation	Testing completion	All testing phases completed with sign-offs.
	Security compliance	Security audit passed with no critical vulnerabilities.
	User acceptance	UAT approved by stakeholders.
VI. Timeline Adherence	Project Completion Date	Completed by **May 31, 2026** (6 months).
	Key Launch Milestones	Soft launch by May 11, 2026; Full public launch by May 25, 2026.
VII. Budget Compliance	Total Project Cost	Within approved budget (PKR 5.06 million).

	Variance	Less than ±5%.
VIII. Stakeholder Satisfaction	Management Approval	Management approval and sign-off.
	End-User Feedback	Positive feedback from end users (Net Promoter Score 70+).
	Process Validation	Expert validation of operational workflows.

Scope Change Management

Any requests to add features, modify deliverables, or change project boundaries must go through formal change control process:

1. **Change Request Submission:**
    - Requestor documents proposed change with justification
    - Impact analysis conducted (timeline, budget, resources, risks)
2. **Change Review:**
    - Project Manager and Technical Lead assess feasibility
    - Stakeholder committee reviews business value vs. cost
3. **Approval Process:**
    - Minor changes (<1 week effort): Project Manager approval
    - Major changes: Stakeholder committee approval required
4. **Implementation:**
    - Approved changes integrated into project plan
    - Team notified and work scheduled
    - Documentation updated

Scope Creep Prevention:

- Weekly scope reviews during development
- Clear acceptance criteria for all deliverables
- "Nice to have" features explicitly logged for Phase 2
- Regular stakeholder communication on scope boundaries

Software Process Model

Selected Model: Agile Development with Scrum Framework

The FoodHub project will utilize an Agile development methodology following the Scrum framework. This iterative and incremental approach is ideal for the project's requirements and constraints.

Justification for Agile/Scrum:

1. **Flexible Requirements:** Food delivery systems require frequent adjustments based on user feedback, market conditions, and operational realities. Agile allows the team to adapt requirements throughout development rather than being locked into initial specifications.

2. **Rapid Time-to-Market:** The 6-month timeline demands quick delivery of working features. Agile's sprint-based approach ensures usable increments are delivered every 2 weeks, allowing early testing and validation.

3. **Stakeholder Engagement:** Restaurant management, delivery operations staff, and the expert consultant need regular involvement to validate solutions. Agile's sprint reviews and demos facilitate continuous stakeholder feedback.

4. **Risk Mitigation:** Building and testing features incrementally identifies technical and operational issues early, reducing the risk of major failures late in the project.

5. **Cross-Functional Collaboration:** The diverse team (developers, designers, QA, operations managers) benefits from Agile's emphasis on collaboration, daily communication, and shared ownership.

Assumptions, Constraints and Dependencies

Project Assumptions

Category	Assumption (Condition Accepted as True)
Technical	Alvigha has reliable high-speed internet (20+ Mbps) at Gulshan branch with backup connectivity.

	Delivery riders have Android smartphones (8.0+) capable of 8+ hour GPS operation with power banks.
	Cloud hosting providers maintain 99.9%+ uptime throughout the project.
	Third-party APIs (Google Maps, payment gateways, SMS) remain available with stable pricing.
	Development team has access to required tools, IDEs, testing devices, and licenses.
Operational	Sufficient qualified candidates available to recruit 20 delivery riders within planned timeline.
	Riders own motorcycles/scooters or can lease vehicles through third-party partners within budget.
	Kitchen can handle increased order volume (150+ orders/day) without compromising quality.
	Restaurant staff will adopt new digital system with proper training.
	Customers in Gulshan willing to download and use new app with competitive pricing.
Business	Management provides necessary approvals, funding, and decision-making support throughout 6 months.
	Expert consultant available for regular consultations during development and operations setup.
	Third-party platforms won't dramatically change commission structures or engage in predatory pricing.
	Pakistan's food delivery, e-commerce, and payment regulations remain stable.
	Customer demand for delivery in Gulshan remains stable or grows, supporting 4,500 monthly orders.
Financial	Full project budget (PKR 5,060,000) secured and available for disbursement per milestones.

	Payment gateway fees remain at approximately 2.5-3% of transaction value.
	Monthly operational costs won't exceed projections by more than 10%.

Project Constraints

Time Constraints:

1. **Fixed 6-Month Deadline:** Must launch by May 31, 2026, for new branch opening
2. **Rider Training:** Only 3 days allocated, limiting training depth
3. **Staff Training:** 2 days for restaurant staff, requiring efficient programs
4. **App Store Approval:** 2-7 days approval process creates timeline uncertainty

Resource Constraints:

5. **Team Size:** Limited to 13 internal resources due to budget
6. **Development Devices:** Limited to 10 physical devices for testing
7. **Rider Fleet:** Initial fleet limited to 20 riders due to equipment budget
8. **Expert Availability:** Maximum 10 hours per week due to other commitments

Financial Constraints:

9. **Fixed Budget:** PKR 5,060,000 cap (including 15% contingency)
10. **Procurement Budget:** PKR 400,000 for delivery equipment
11. **Marketing Budget:** PKR 200,000 for launch campaign
12. **Monthly Operations:** PKR 275,000 cap per month for first 6 months

Technical Constraints:

13. **Technology Stack:** Must use familiar technologies (no new languages/frameworks)
14. **Mobile Platforms:** iOS 13+ and Android 8.0+ only (older devices in Phase 2)
15. **API Rate Limits:** Google Maps free tier limited to 28,500 requests/month
16. **SMS Limits:** Free tier limited to 1,000 messages/month (PKR 1.50 per additional)
17. **Cloud Infrastructure:** Sized for 500 concurrent users initially

Business and Legal Constraints:

18. **Single Location:** Platform serves Gulshan branch only initially
19. **Payment Compliance:** Must maintain PCI-DSS compliance requirements

20. **Labor Laws:** Rider classification affects legal obligations and costs
21. **Data Protection:** Must comply with Pakistan data protection regulations
22. **Food Safety:** Must comply with food safety and handling regulations

Operational Constraints:

23. **Delivery Radius:** 5km limit from restaurant (food quality and delivery time)
24. **Operating Hours:** 11 AM - 11 PM daily (no 24/7 in Phase 1)
25. **Kitchen Capacity:** Maximum 20 concurrent orders can be prepared
26. **Customer Support:** Available only during operational hours (11 AM - 11 PM)

External Dependencies

Dependency Type	Component (Priority)	Dependency/Risk	Mitigation Strategy
Critical Third-Party Services	**Payment Gateway Providers** (Critical)	Risk of service outages, API changes, or pricing increases (Stripe, JazzCash, EasyPaisa APIs).	Use multiple providers for redundancy; offer **Cash on Delivery (COD)** backup; implement service monitoring.
	Google Maps Platform (Critical)	Risk of API unavailability, rate limit exceedance, or pricing changes (Geocoding, Routing).	Utilize **location caching**; implement usage monitoring; include a budget buffer for overages.
	Cloud Hosting Provider (Critical)	Risk of major service outages resulting in complete downtime (AWS/Google Cloud).	Employ **Multi-Availability Zone deployment**; ensure automated backups and disaster recovery plans; use 24/7 monitoring.
High Third-Party Services	**SMS Gateway Service** (High)	Risk of outages or message delivery failures (Twilio or local provider).	Provide **email and push notification alternatives**; monitor delivery status.
	Firebase Cloud Messaging (FCM) (High)	Risk of service downtime preventing real-time updates.	Implement **SMS fallback** for critical notifications; ensure email backup.

	App Store Platforms (High)	Risk of app rejection, delayed approval, or policy changes (Apple/Google).	**Submit 2 weeks early**; ensure full compliance with guidelines; have a resubmission contingency plan.
Regulatory	**Business Licensing Authorities** (High)	Risk of delayed approvals for e-commerce and food delivery licenses.	**Start applications in Month 1**; engage a legal consultant; ensure complete documentation.
	Payment Regulatory Bodies (Medium)	Risk of non-compliance restricting payment processing (State Bank of Pakistan).	Use **licensed gateways** that are already compliant; maintain meticulous transaction records and audits.
Market	**Customer Adoption Rate** (Critical)	Risk of low adoption, failing revenue projections and ROI.	Implement **aggressive promotions** (25% off); utilize referral programs and influencer marketing.
	Rider Recruitment Market (High)	Risk of insufficient qualified candidates limiting delivery capacity.	Offer **competitive compensation**; begin recruitment 2 months early; explore employment agency partnerships.
	Competitive Landscape (Medium)	Risk that competitors slash commissions or prices, weakening FoodHub's value proposition.	Focus on **unique value** (exclusive items, loyalty, faster delivery); maintain agile pricing strategies.
Operational	**Internet Service Provider (ISP)** (Critical)	Risk of reliable internet loss at the restaurant, halting order intake.	Provide a **backup 4G LTE hotspot**; implement an offline order queue; utilize UPS backup.
	Restaurant Kitchen Operations (Critical)	Risk of the kitchen failing to maintain quality/speed with increased volume, leading to bottlenecks.	Implement **workflow optimization** and staff training; phase volume increase; use real-time monitoring.

	Electricity Supply (Medium)	Risk of frequent power outages disrupting restaurant equipment and systems.	Install **UPS backup systems**; utilize a generator for extended outages.

Stakeholders

Stakeholder Identification and Roles

Primary Stakeholders:

Stakeholder Group	Role & Representatives	Key Interests	Level of Influence
1. Executive Management	Project Sponsor (CEO/Owner, CFO, COO)	ROI, market positioning, brand reputation, and scalability.	**Very High** (Final authority on budget, scope, and strategic decisions).
2. Customers	End Users of the mobile application (Target: 5,000+ users).	User-friendly app, fast delivery, competitive prices, quality food, and responsive support.	**Very High** (Revenue source and word-of-mouth marketing).
3. Delivery Riders	Front-line delivery staff (20 riders).	Easy app, reliable navigation, fair compensation, safety, and clear performance metrics.	**High** (Direct impact on service quality and customer satisfaction).
4. Project Manager	Overall project coordination and delivery.	On-time/budget/scope delivery, team coordination, and stakeholder satisfaction.	**High** (Day-to-day execution authority).
5. Development Team	Technical staff (Devs, Designers, QA, BA, Ops Manager, etc. - 11 members total).	Clear requirements, realistic timelines, overcoming technical challenges, and recognition.	**High** (Direct impact on product quality and delivery speed).

6. Restaurant Branch Manager	Operational lead for restaurant-side implementation.	Smooth integration, staff adoption, efficient workflow impact, and performance metrics.	**High** (Daily operational decisions and staff management).
7. Expert Consultant	Provides domain expertise and operational workflow validation.	Providing industry insights, ensuring practical solutions, and professional reputation.	**Medium** (Advisory capacity, no decision authority).
8. Kitchen Staff	End users of the order management system.	Easy-to-use system, minimal workflow disruption, training, and reliability.	**Medium** (Adoption is critical for operational success).
9. Front-Desk Staff	Customer query and order issue support.	Simple interface, quick order access, clear refund processes, and training.	**Medium** (Direct customer interaction).

Secondary Stakeholders:

Stakeholder Group	Role in Project	Key Interests	Level of Influence
Payment Gateway Providers	Enable secure payment processing.	Transaction volume and processing fee revenue.	**Medium** (Critical service, but limited involvement in development).
Marketing Team/Agency	Launch marketing and customer acquisition.	Campaign success metrics and brand visibility.	**Medium** (Effectiveness impacts adoption).
Legal and Compliance Advisors	Regulatory compliance and risk management.	Legal compliance, risk mitigation, and contract review.	**Medium** (Advisory on legal matters).
Equipment/Uniform Suppliers	Provide delivery bags, uniforms, and vehicle branding.	Timely delivery of quality materials within budget.	**Low** (Vendors fulfilling contracts).
Vehicle Leasing Companies	Provide motorcycles/scooters for riders.	Fleet utilization and lease payments.	**Low** (Service providers with contractual obligations).
Competitors	External market forces influencing expectations.	Monitoring competitive positioning.	**Low** (No direct involvement, but market influence).

Stakeholder Responsibilities for Project Aspects

Stakeholder Group	Planning Responsibilities	Execution/Development Responsibilities	Launch & Post-Launch Duties	Signing Authority For
Executive Management	Approve project charter, budget, strategic direction, feasibility analysis.	Monthly steering meetings, milestone reviews, scope change approvals, roadblock removal.	Go-live decision approval, marketing campaign support, organizational communication.	**Project Charter, Budget Approvals, Major Scope Changes, Final Go-Live Decision**
Project Manager	Develop project plan (WBS/timeline), risk assessment, communication plan, requirements coordination.	Daily stand-ups, progress monitoring, risk/issue management, vendor coordination, weekly status reports.	Oversee deployment, go-live readiness assessment, post-launch stabilization, lessons learned.	**Weekly Status Reports, Vendor Contracts** ($<$PKR 100K), **Team Work Assignments, QA Reports**
Restaurant Branch Manager	Current workflow insights, pain point identification, staff allocation approval.	Staff training coordination, portal usability feedback, testing facilitation, restaurant readiness.	Training completion oversight, operational integration monitoring, staff adoption support.	**Staff Training Completion, Restaurant Portal UAT, Operational Readiness**
Technical Leads	System architecture design, integration oversight.	Code reviews, technical decisions, blocker resolution.	N/A	**Technical Design Documents, Code Review Approvals, System Integration Tests**
Frontend Developers	N/A	Mobile app development (customer/rider), UI/UX implementation, backend API integration, device compatibility testing.	N/A	N/A
Backend Developers	Database schema design.	RESTful API and business logic development, third-party integrations (payments, maps,	N/A	N/A

		SMS), security and authentication systems.		
UI/UX Designers	User research, wireframing, mockup creation, visual design system, responsive design specs.	Usability testing.	N/A	N/A
Business Analyst	Requirements gathering and documentation, user story creation with acceptance criteria.	Stakeholder validation, UAT coordination.	N/A	N/A
QA Tester	Test plan/case creation, functional/performance/security testing.	Bug documentation, regression testing.	N/A	**Test Reports, UAT Results**
Database Administrator	N/A	Database schema optimization, backup/recovery procedures, performance tuning, data security, query optimization.	N/A	N/A
Delivery Ops Manager	Delivery zone mapping, route optimization strategies, SLA definition, rider training program design, operational workflow definition.	N/A	N/A	N/A
Rider Coordinator	N/A	Rider recruitment/onboarding, training delivery, shift scheduling, performance metrics definition, rider support coordination.	N/A	N/A
Marketing Team	Launch marketing strategy, promotional campaign design.	Marketing collateral creation (social media, ads, posters), app store optimization materials.	Campaign execution, performance monitoring, social	**Marketing Materials, Campaign Budgets, Promotional Offers**

			media management, customer engagement.	
Restaurant Staff	Requirement gathering participation.	Usability testing feedback, training attendance.	New system adoption, SOP compliance, issue reporting, continuous improvement feedback.	N/A
Customers	N/A	N/A	App adoption, usage feedback, order/delivery ratings, satisfaction surveys, reviews/testimonials.	N/A
Payment Gateway Providers	N/A	API documentation and developer support, merchant account approval, security compliance reviews.	Transaction processing monitoring, issue resolution, reconciliation reports.	N/A

Deliverable Responsibilities Matrix

Deliverable	Creator	Reviewer	Approver	Signing Authority
Project Charter	Project Manager	Dev Team Leads	CEO/Owner	CEO/Owner
Project Plan & WBS	PM + Expert	Dev Team	Branch Mgr, COO	COO
Feasibility Analysis	BA + PM	Expert Consultant	CFO	CFO
Requirements Doc	Business Analyst	All Stakeholders	PM, Branch Mgr	Branch Manager
Technical Architecture	Technical Leads	Dev Team	PM	Project Manager
UI/UX Designs	UI/UX Designers	BA, Restaurant Staff	PM, Branch Mgr	Project Manager
Customer Mobile App	Frontend Devs	QA	PM, Customers	Project Manager
Rider Mobile App	Frontend Devs	QA, Rider Coord	PM, Pilot Riders	Project Manager

Restaurant Portal	Frontend Devs	QA, Staff	Branch Manager	Branch Manager
Admin Dashboard	Frontend Devs	QA, PM	COO	COO
Backend APIs	Backend Devs	Tech Leads, QA	PM	Technical Lead
Database System	DBA	Backend Devs	PM	DBA
Cloud Infrastructure	Backend Devs, DBA	Tech Leads	PM	Technical Lead
Payment Integration	Backend Devs	QA, Finance	CFO	CFO
Test Reports	QA	Tech Leads	PM	QA Tester
Security Audit	External Auditor	DBA, Tech Leads	CFO, PM	CEO/Owner
Training Materials	Ops Mgr, Rider Coord, BA	Staff, Riders	Branch Mgr	Branch Manager
Training Completion	Branch Mgr, Rider Coord	PM	COO	COO
Delivery Equipment	Procurement	Rider Coord	Branch Mgr	Branch Manager
Marketing Materials	Marketing Team	Branch Mgr, PM	CEO/Owner	CEO/Owner
Launch Readiness	PM	All Leads, Branch Mgr	CEO/Owner	CEO/Owner
Go-Live Decision	PM + COO	Executive Mgmt	CEO/Owner	CEO/Owner
Post-Launch Review	PM	All Stakeholders	Executive Mgmt	CEO/Owner

Expert

Detail	Description
Expert Chosen	Muhammad Ali
Domain	Product & Project Management, E-commerce Systems, Digital Strategy, Agile Delivery.
Job Title	Senior Project Manager / Product Manager (E-commerce Systems)
Contact	Muhammad Ali - Linkedin
Availability Constraint	Maximum 10 hours per week (as per project constraints)

Justification for Selection (Expert's Statement)

"I have been selected as an expert because I have led multiple full-cycle e-commerce product launches at Dvago.pk, managing diverse teams such as UI/UX, data analytics, engineering, QA, DevOps, and security. I have hands-on experience building scalable systems, integrating payment gateways, and aligning product roadmaps with business needs. My background allows me to guide this project in terms of feasibility, resource planning, and risk mitigation."

Past Experiences & Successes

Product/Area	Key Achievements & Expertise
Digital Product Launches	Successfully launched more than five major digital products including an Order Management System, Inventory Management System, Logistics Management System, and multiple B2B & B2C e-commerce platforms.
Third-Party Integrations	Led integrations with Meezan Bank, UBL, Savyour, Vouch365, Mixpanel, GA4, WebEngage, and e.ocean - enhancing both functionality and user experience.
Core Strengths	PRD/BRD development, backlog ownership, cross-functional team leadership, and ensuring timely delivery.

Role in the FoodHub Project (Expert's Statement)

"In this project, my role is to provide expert guidance on structuring the team, planning the schedule, validating the WBS, ensuring technical feasibility, defining requirements, and reviewing integration-related and timeline-related decisions. I also validate task dependencies, resource allocation, and major risks."

Work Breakdown Structure (WBS)

1.0 PROJECT INITIATION & PLANNING (Weeks 1-2)

1.1 Project Initiation (Activities 1-6)

- 1.1.1 Conduct project kickoff meeting with stakeholders
- 1.1.2 Define and approve project charter
- 1.1.3 Establish project governance and communication structure
- 1.1.4 Conduct stakeholder identification and create register
- 1.1.5 Schedule expert consultant sessions
- 1.1.6 Set up project repository and documentation tools

1.2 Requirements Gathering (Activities 7-13)

- 1.2.1 Conduct customer needs analysis workshop
- 1.2.2 Interview restaurant staff for operational requirements

- 1.2.3 Gather delivery rider requirements
- 1.2.4 Document functional requirements for all systems
- 1.2.5 Create use case diagrams for all user types
- 1.2.6 Validate requirements with stakeholders
- 1.2.7 Finalize requirements document with expert approval

1.3 Technical Planning (Activities 14-20)

- 1.3.1 Evaluate and select technology stack
- 1.3.2 Design high-level system architecture
- 1.3.3 Select cloud hosting provider and services
- 1.3.4 Select payment and SMS gateway providers
- 1.3.5 Set up development environments
- 1.3.6 Establish version control and CI/CD pipeline
- 1.3.7 Finalize project plan with WBS and timeline

2.0 DESIGN PHASE (Weeks 3-6)

2.1 UI/UX Design - Customer App (Activities 21-28)

- 2.1.1 Conduct user research and create personas
- 2.1.2 Create user journey maps and information architecture
- 2.1.3 Design wireframes for all customer app screens
- 2.1.4 Design high-fidelity mockups for home and menu screens
- 2.1.5 Design shopping cart and checkout flow
- 2.1.6 Design order tracking interface with map
- 2.1.7 Design profile and order history screens
- 2.1.8 Conduct usability testing with target users

2.2 UI/UX Design - Rider App (Activities 29-34)

- 2.2.1 Create rider personas and journey maps
- 2.2.2 Design rider dashboard and order assignment interface
- 2.2.3 Design navigation and routing screens
- 2.2.4 Design delivery confirmation flow
- 2.2.5 Design earnings and performance dashboard
- 2.2.6 Validate designs with delivery riders

2.3 UI/UX Design - Restaurant Portal (Activities 35-40)

- 2.3.1 Design restaurant dashboard and order queue
- 2.3.2 Design kitchen display system (KDS) interface
- 2.3.3 Design menu and inventory management screens
- 2.3.4 Design sales reporting interface
- 2.3.5 Validate designs with restaurant staff

- 2.3.6 Create design system and component library

2.4 UI/UX Design - Admin Dashboard (Activities 41-44)

- 2.4.1 Design admin dashboard with key metrics
- 2.4.2 Design analytics and reporting interfaces
- 2.4.3 Design user and system management screens
- 2.4.4 Design promotional campaign management interface

2.5 Database Design (Activities 45-52)

- 2.5.1 Create entity-relationship diagram (ERD)
- 2.5.2 Design user, authentication, and profile tables
- 2.5.3 Design menu, category, and item tables
- 2.5.4 Design order and order items tables
- 2.5.5 Design payment and transaction tables
- 2.5.6 Design delivery, rider, and tracking tables
- 2.5.7 Design loyalty program and promotional tables
- 2.5.8 Normalize schema and create indexing strategy

3.0 DEVELOPMENT PHASE (Weeks 7-16)

3.1 Backend Development - Foundation (Activities 53-65)

- 3.1.1 Set up Node.js backend framework and structure
- 3.1.2 Configure PostgreSQL database with migrations
- 3.1.3 Set up Redis caching layer
- 3.1.4 Implement JWT authentication and authorization
- 3.1.5 Implement user registration and OTP verification
- 3.1.6 Implement password reset functionality
- 3.1.7 Implement role-based access control (RBAC)
- 3.1.8 Set up API rate limiting and error handling
- 3.1.9 Implement logging and monitoring services
- 3.1.10 Set up API documentation (Swagger/OpenAPI)
- 3.1.11 Create automated database backup system
- 3.1.12 Implement security middleware and validation
- 3.1.13 Set up testing framework and initial tests

3.2 Backend Development - Core Features (Activities 66-82)

- 3.2.1 Implement menu management APIs (CRUD)
- 3.2.2 Implement shopping cart APIs
- 3.2.3 Implement order placement and management APIs
- 3.2.4 Implement order status update workflow
- 3.2.5 Implement real-time tracking with WebSocket

- 3.2.6 Integrate Stripe payment gateway
- 3.2.7 Integrate JazzCash payment API
- 3.2.8 Integrate EasyPaisa payment API
- 3.2.9 Implement payment webhook handlers
- 3.2.10 Integrate Google Maps API (geocoding, routing)
- 3.2.11 Implement rider assignment algorithm
- 3.2.12 Implement rider location tracking API
- 3.2.13 Integrate Firebase Cloud Messaging (FCM)
- 3.2.14 Integrate Twilio SMS service
- 3.2.15 Implement notification scheduling service
- 3.2.16 Implement loyalty points calculation engine
- 3.2.17 Implement promotional code validation API

3.3 Frontend Development - Customer Mobile App (Activities 83-97)

- 3.3.1 Set up React Native project structure
- 3.3.2 Implement splash screen and onboarding
- 3.3.3 Implement user registration and login flow
- 3.3.4 Implement home screen with promotions
- 3.3.5 Implement menu browsing with categories
- 3.3.6 Implement search and filtering functionality
- 3.3.7 Implement product detail and customization
- 3.3.8 Implement shopping cart management
- 3.3.9 Implement checkout and payment integration
- 3.3.10 Implement order tracking with live map
- 3.3.11 Implement push notification handling
- 3.3.12 Implement order history and reordering
- 3.3.13 Implement user profile management
- 3.3.14 Implement ratings and review system
- 3.3.15 Optimize performance and test on devices

3.4 Frontend Development - Rider Mobile App (Activities 98-107)

- 3.4.1 Set up Android project for rider app
- 3.4.2 Implement rider authentication and profile
- 3.4.3 Implement order list and assignment screen
- 3.4.4 Implement order acceptance/rejection flow
- 3.4.5 Implement Google Maps navigation integration
- 3.4.6 Implement background GPS tracking service
- 3.4.7 Implement customer communication (call/message)
- 3.4.8 Implement delivery confirmation with PIN
- 3.4.9 Implement earnings and performance dashboard

- 3.4.10 Implement availability toggle and optimize battery

3.5 Frontend Development - Restaurant Portal (Activities 108-114)

- 3.5.1 Set up React.js project for restaurant portal
- 3.5.2 Implement staff authentication system
- 3.5.3 Implement real-time order queue with alerts
- 3.5.4 Implement order details and status updates
- 3.5.5 Implement menu management interface
- 3.5.6 Implement inventory management
- 3.5.7 Implement sales reports and analytics

3.6 Frontend Development - Admin Dashboard (Activities 115-120)

- 3.6.1 Set up admin dashboard project structure
- 3.6.2 Implement admin authentication and authorization
- 3.6.3 Implement operational dashboard with real-time metrics
- 3.6.4 Implement analytics and reporting visualizations
- 3.6.5 Implement user management (customers, riders, staff)
- 3.6.6 Implement promotional campaign and system configuration

4.0 TESTING & QUALITY ASSURANCE (Weeks 17-20)

4.1 Test Planning (Activities 121-123)

- 4.1.1 Create comprehensive test plan and strategy
- 4.1.2 Define test scenarios and create test cases
- 4.1.3 Set up test environment and prepare test data

4.2 Functional Testing (Activities 124-131)

- 4.2.1 Execute customer app functional tests
- 4.2.2 Execute rider app functional tests
- 4.2.3 Execute restaurant portal functional tests
- 4.2.4 Execute admin dashboard functional tests
- 4.2.5 Test payment gateway integrations end-to-end
- 4.2.6 Test notification services (SMS and push)
- 4.2.7 Test complete order workflow from placement to delivery
- 4.2.8 Document defects and execute regression testing

4.3 Performance & Security Testing (Activities 132-137)

- 4.3.1 Conduct load testing (200 concurrent users)
- 4.3.2 Conduct stress testing (500+ concurrent users)
- 4.3.3 Optimize API response times and database queries
- 4.3.4 Conduct security vulnerability scanning

- 4.3.5 Perform penetration testing on all systems
- 4.3.6 Address all critical security vulnerabilities

4.4 Compatibility & UAT (Activities 138-142)

- 4.4.1 Test customer app on 10+ device models
- 4.4.2 Test rider app on 5+ Android devices
- 4.4.3 Test web portals across multiple browsers
- 4.4.4 Conduct user acceptance testing (UAT) with stakeholders
- 4.4.5 Obtain UAT sign-off and approval

5.0 DEPLOYMENT & INFRASTRUCTURE (Weeks 21-22)

5.1 Cloud Infrastructure Setup (Activities 143-150)

- 5.1.1 Provision production cloud servers (AWS/GCP)
- 5.1.2 Configure load balancer and auto-scaling
- 5.1.3 Set up production database with replication
- 5.1.4 Configure Redis cache cluster
- 5.1.5 Set up CDN for static assets
- 5.1.6 Install SSL certificates and configure HTTPS
- 5.1.7 Set up monitoring, alerting, and logging systems
- 5.1.8 Configure automated backup and disaster recovery

5.2 Application Deployment (Activities 151-157)

- 5.2.1 Deploy backend APIs to production
- 5.2.2 Deploy admin dashboard to production
- 5.2.3 Deploy restaurant portal to production
- 5.2.4 Submit and publish customer app to App Store
- 5.2.5 Submit and publish customer app to Google Play
- 5.2.6 Submit and publish rider app to Google Play
- 5.2.7 Conduct production smoke testing and validation

6.0 OPERATIONAL SETUP (Weeks 21-22)

6.1 Delivery Operations (Activities 158-163)

- 6.1.1 Define delivery zones and configure pricing
- 6.1.2 Procure delivery bags, uniforms, and equipment
- 6.1.3 Recruit and interview 20 delivery riders
- 6.1.4 Complete rider onboarding and documentation
- 6.1.5 Distribute equipment and uniforms to riders
- 6.1.6 Set up rider scheduling and zone assignments

6.2 Training Programs (Activities 164-169)

- 6.2.1 Develop training materials for all user types
- 6.2.2 Create video tutorials and documentation
- 6.2.3 Conduct 2-day restaurant staff training workshop
- 6.2.4 Conduct 3-day delivery rider training program
- 6.2.5 Conduct admin dashboard training for management
- 6.2.6 Evaluate training effectiveness with assessments

6.3 Marketing & Launch Prep (Activities 170-174)

- 6.3.1 Develop brand identity and marketing guidelines
- 6.3.2 Create social media content and campaigns
- 6.3.3 Design in-store promotional materials
- 6.3.4 Set up launch promotions (25% off, referral program)
- 6.3.5 Configure loyalty program in system

7.0 LAUNCH & STABILIZATION (Weeks 23-26)

7.1 Soft Launch (Activities 175-178)

- 7.1.1 Launch with limited delivery zone (Week 23)
- 7.1.2 Monitor system performance 24/7
- 7.1.3 Collect user feedback and address issues
- 7.1.4 Optimize workflows based on real operations

7.2 Full Launch (Activities 179-182)

- 7.2.1 Execute full marketing campaign
- 7.2.2 Expand delivery zones progressively
- 7.2.3 Scale rider fleet based on demand
- 7.2.4 Monitor KPIs and customer acquisition

7.3 Project Closure (Activities 183-186)

- 7.3.1 Conduct post-launch performance review
- 7.3.2 Document lessons learned
- 7.3.3 Create project closure report
- 7.3.4 Handover to operations team with final presentation

Expert Notes on WBS (Muhammad Ali)

"I have reviewed and approved the 180+ activity WBS. The structure, numbering, and phases are accurate and cover all major areas such as requirements, design, development, integrations, testing, and deployment."

Expert Verification on MS Project Plan(Muhammad Ali)

"I have verified the task durations, start-end dates, and predecessor logic. Earlier, some predecessor dependencies were misaligned - especially within the integration and testing phases. These have now been corrected to ensure realistic sequencing and avoid parallel tasks that depend on each other. The updated schedule now reflects feasible development flow and eliminates circular dependencies."

Risk Management

RISK #1: CRITICAL SYSTEM INTEGRATION & TECHNOLOGY INEXPERIENCE

Category: Technical / HR / Scope
Probability: High | Impact: Very High

The project requires integration of multiple systems including:
• Stripe, JazzCash, EasyPaisa (Activities 3.2.6–3.2.9)
• WebSocket real-time tracking (Activity 3.2.5)
• Google Maps routing (Activity 3.2.10)
• Firebase Cloud Messaging (Activity 3.2.13)
• Twilio SMS (Activity 3.2.14)

These require expertise in authentication flows, webhook logic, sandbox testing, and performance optimization.
If the team lacks experience, Activities 3.1.1–3.1.13 and 66–82 may take significantly longer, causing delays and architectural issues.

Impact:
• Integration failures stop frontend progress
• Unstable real-time features
• Late-stage security issues
• Timeline instability

Consultation by expert: Integrating payment gateways, tracking APIs, and SMS/OTP services is high-risk due to vendor dependence, especially with limited team experience in real-time tracking and cloud technologies.

Strategy:
• Early integration testing
• Sandbox setup for all third-party services
• Senior code reviews
• API contract finalization (Activity 3.1.10)

RISK #2: INACCURATE TIME ESTIMATES & TEAM AVAILABILITY
Category: Schedule / HR
Probability: High | Impact: Very High

Backend development (Activities 53–82) includes authentication, rider assignment, payment integration, and monitoring.
These activities realistically require more than the allocated 10 weeks.
Underperforming or unavailable members may delay dependent tasks.

Impact:
• Frontend delays (Activities 83–120)
• Testing compression (Activities 121–142)
• Deployment delays

Consultation by expert: New features, unclear requirements, and testing dependencies often lead to incorrect timelines.

Strategy:
• Senior-led re-estimation
• Add buffer of ~25%
• Weekly sprint reviews
• Daily standups

RISK #3: SCOPE CREEP LEADING TO QUALITY DEGRADATION
Category: Scope & Quality
Probability: High | Impact: High

FoodHub has diverse stakeholders - all requesting additional features once development begins (analytics for restaurants, earnings tools for riders, social/loyalty features for customers, AI features for management).
Uncontrolled additions cause scope expansion and directly threaten product quality, especially when features are rushed into an architecture not designed for them.

Impact:
- Quality Impacts:
    - Increased defects due to incomplete testing
    - Technical debt and unstable integrations
    - Performance issues and inconsistent UX
    - Higher post-launch failure risk

- Scope Impacts:
    - Timeline and budget become unreliable
    - MVP focus is lost
    - Repeated launch delays
    - Reduced team motivation and stakeholder confidence

Consultation by expert: FoodHub involves many stakeholders, and uncontrolled feature additions threaten both quality and deadlines.

Strategy:

- Define & communicate MVP clearly in the project charter; get stakeholder sign-off.
- Strict change control: any new request requires time, cost, and quality impact analysis.
- Maintain a Phase 2 backlog to defer non-MVP features without rejecting stakeholder ideas.
- Scope & quality validation reviews at major milestones (Weeks 6, 16, 20, 22).
- Educate stakeholders on how late changes compromise stability and testing.

Decision rule: new features must replace an existing one or extend launch timeline - no "quick adds."

RISK #4: POOR COST ESTIMATES FOR CLOUD INFRASTRUCTURE
Category: Cost / Budget
Probability: Medium | Impact: High

Cloud cost factors include:
• Google Maps API ($5–$7 per 1,000 calls)
• Twilio SMS (Rs. 1–2 per SMS)
• Payment gateway fees (2–3%)
• Auto-scaling costs based on peak hours
• Backup storage (Activity 5.1.8)
• Monitoring & logging (Activity 5.1.6)

Consultation by expert: Dynamic rider tracking and real-time order updates can spike compute, storage, and bandwidth charges.

Strategy:
• Cost modelling for 100–1000 users
• Billing alerts from Day 1
• Caching with Redis (Activity 5.1.4)
• CDN usage (Activity 5.1.5)

RISK #5: RIDER RECRUITMENT & RETENTION CHALLENGES
Category: HR / Operations
Probability: High | Impact: High

Activities 6.1.3–6.1.4 require hiring 20 riders in 2 weeks.
Industry turnover is 40–50% in the first 3 months.
Competition and income uncertainty create further challenges.

Impact:
• Late deliveries
• Poor customer experience during soft launch (Week 23)
• Burnout of existing riders

Consultation by expert: The platform heavily depends on a reliable delivery fleet; recruitment shortages can impact operations.

Strategy:
• Begin recruitment earlier (Week 12)
• Hire 30 riders anticipating attrition
• Training program (Activity 6.2.4)
• Referral bonuses (Activity 6.3.4)
• Rider support hotline